

Academic Management System - Design Documentation

This document integrates all design artifacts of the **Academic Management System (AMS)**. It includes:

- **Software Requirements Specification (SRS)**
- **Data Dictionary**
- **Entity-Relationship Diagram (ERD)**
- **Data Flow Diagram (DFD)**
- **UML Class Diagram**

The goal is to provide a comprehensive view of AMS, from requirements to design.

1. Software Requirements Specification (SRS)

1.1 Purpose

The Academic Management System (AMS) aims to support student enrollment, course and term management, grade recording, attendance tracking, and reporting. It is designed for three primary user groups: **Students**, **Teachers**, and **Administrators**.

1.2 Scope

- **Students:** enroll in courses, view transcripts, check attendance.
- **Teachers:** manage course offerings, record grades, record attendance.
- **Administrators:** configure terms, create courses, assign teachers, generate reports.

1.3 Functional Requirements

- Student course enrollment
- Teacher grade and attendance submission
- Administrator management of courses and terms
- Reporting and analytics for all roles

1.4 Non-Functional Requirements

- **Performance:** support at least 500 concurrent users
- **Security:** role-based access control
- **Scalability:** compatible with PostgreSQL and AWS RDS deployment
- **Reliability:** daily backups, recovery support

1.5 Use Cases

Use Case ID	Actor	Description	Preconditions	Postconditions	DFD Mapping	UML Mapping
UC-01	Student	Enroll in a course	Student is logged in	Enrollment record created	Enrollment Manager	Student.enrollCourse()
UC-02	Student	View transcript	Student has completed courses	Transcript report displayed	Reporting & Analytics	Student.viewTranscript() / Reporting.generateTranscript()
UC-03	Student	View attendance	Attendance records exist	Attendance report displayed	Reporting & Analytics	Student.viewAttendance() / Reporting.generateAttendanceReport()
UC-04	Teacher	Record grades for a course offering	Teacher assigned to offering	Grades saved to Grade DB	Grade Manager	Teacher.recordGrade()
UC-05	Teacher	Record attendance for a course offering	Teacher assigned to offering	Attendance saved to Attendance DB	Attendance Manager	Teacher.recordAttendance()
UC-06	Teacher	View class roster	Teacher assigned to offering	Student list displayed	Enrollment Manager	Teacher.viewClassRoster()
UC-07	Admin	Setup academic term and course offerings	Admin privileges granted	Term and course offerings created	Course/Term Manager	Admin.setupCourse()
UC-08	Admin	Assign teacher to a course offering	Teacher exists in system	Teacher assigned to offering	Course/Term Manager	Admin.assignTeacher()
UC-09	Admin	View system-wide statistics and analytics	Admin privileges granted	Reports generated by Reporting module	Reporting & Analytics	Admin.viewStatistics() / Reporting.generateAnalytics()
UC-10	Reporting	Generate transcript, attendance, performance	Relevant data stored in DBs	Reports generated and displayed	Reporting & Analytics	Reporting.generateTranscript(), generateAttendanceReport(), generatePerformanceReport(), generateAnalytics()

2. Data Dictionary

2.1 Student

Attribute	Type	Description
student_id (PK)	int	Unique ID for student
name	varchar(100)	Full name
email	varchar(100)	Unique email address
dob	date	Date of birth
major	varchar(100)	Major field of study
enrollment_year	int	Year of enrollment

2.2 Teacher

Attribute	Type	Description
teacher_id (PK)	int	Unique ID for teacher
name	varchar(100)	Full name
email	varchar(100)	Email
department	varchar(100)	Department
title	varchar(50)	Academic title

2.3 Term

Attribute	Type	Description
term_id (PK)	int	Unique ID for term
year	int	Academic year
semester	varchar(20)	Semester (e.g. Spring, Fall)

2.4 Course

Attribute	Type	Description
course_id (PK)	int	Unique ID for course
code	varchar(20)	Course code (e.g. CS101)
title	varchar(100)	Course title
credits	int	Credit value

2.5 CourseOffering

Attribute	Type	Description
offering_id (PK)	int	Unique ID for offering
section	varchar(10)	Section label (e.g. A, B)
capacity	int	Maximum number of students
term_id (FK)	int	Linked academic term
course_id (FK)	int	Linked course

2.6 OfferingInstructor

Attribute	Type	Description
offering_instructor_id (PK)	int	Unique ID for assignment
offering_id (FK)	int	Linked course offering
teacher_id (FK)	int	Linked teacher

This table manages the many-to-many relationship between CourseOffering and Teacher.

2.7 Enrollment

Attribute	Type	Description
enrollment_id (PK)	int	Unique ID for enrollment
student_id (FK)	int	Linked student
offering_id (FK)	int	Linked course offering
enrollment_date	date	Date of enrollment

2.8 Grade

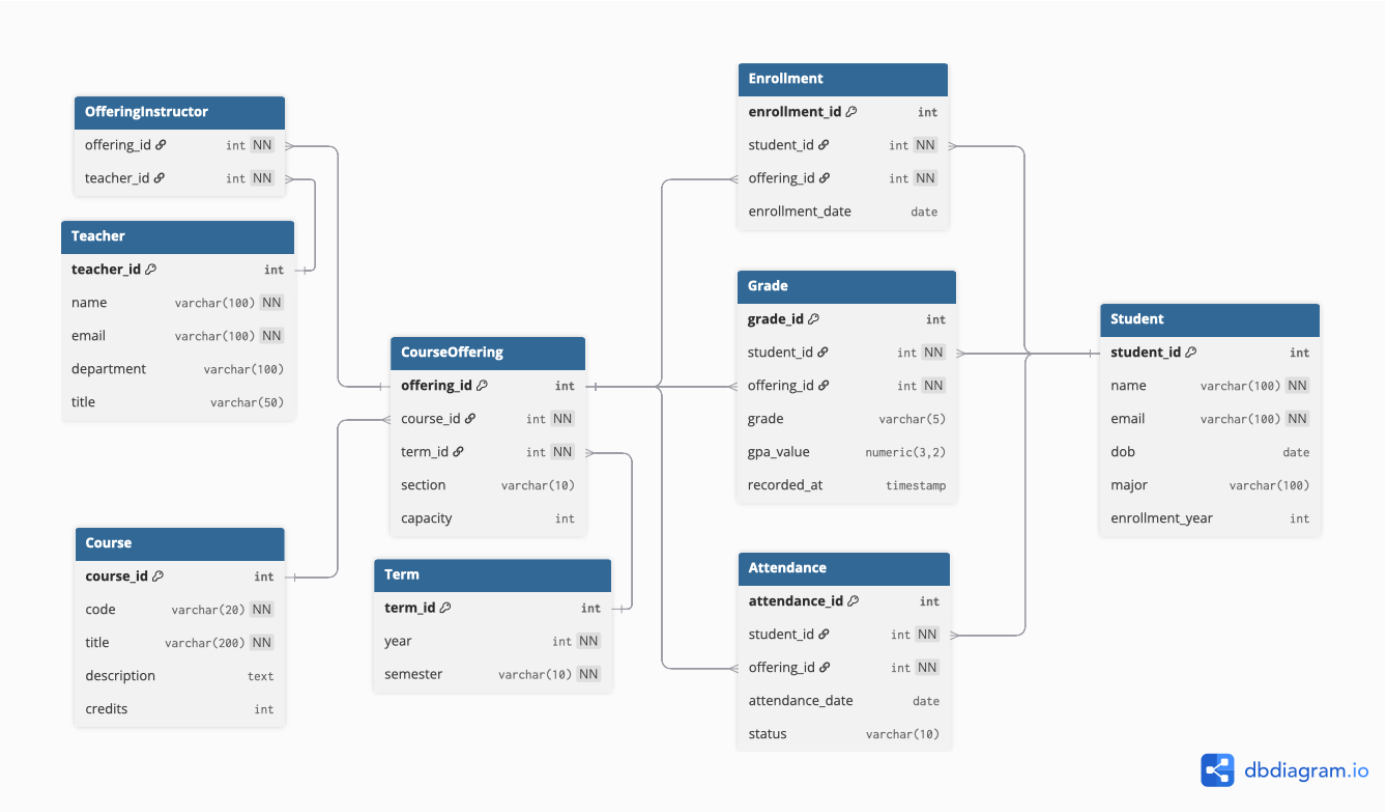
Attribute	Type	Description
grade_id (PK)	int	Unique ID for grade
enrollment_id (FK)	int	Linked enrollment
grade	varchar(5)	Grade value (e.g. A, B, C)
gpa_value	float	GPA equivalent
recorded_at	timestamp	Date/time grade recorded

2.9 Attendance

Attribute	Type	Description
attendance_id (PK)	int	Unique ID for attendance
enrollment_id (FK)	int	Linked enrollment
date	date	Attendance date
status	varchar(10)	Attendance status (Present/Absent)

3. Entity-Relationship Diagram (ERD)

ERD Diagram

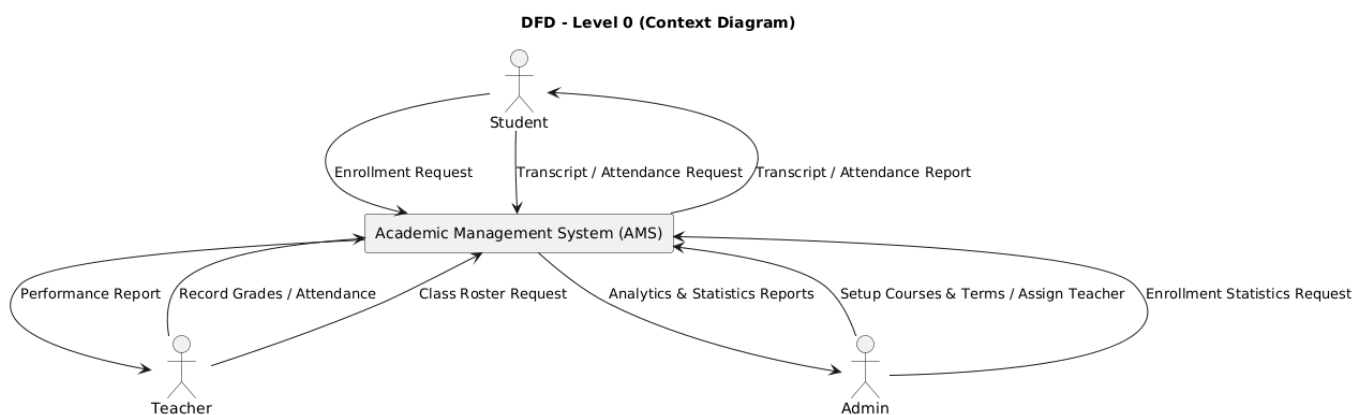


Explanation

- **Student** ↔ **Enrollment** ↔ **CourseOffering** → core link of student course registration
- **Enrollment** ↔ **Grade** → one grade per enrollment
- **Enrollment** ↔ **Attendance** → multiple attendance records per enrollment
- **Course** ↔ **CourseOffering** ↔ **Term** → courses offered within academic terms
- **Teacher** ↔ **CourseOffering** → teachers assigned to course offerings

4. Data Flow Diagram (DFD)

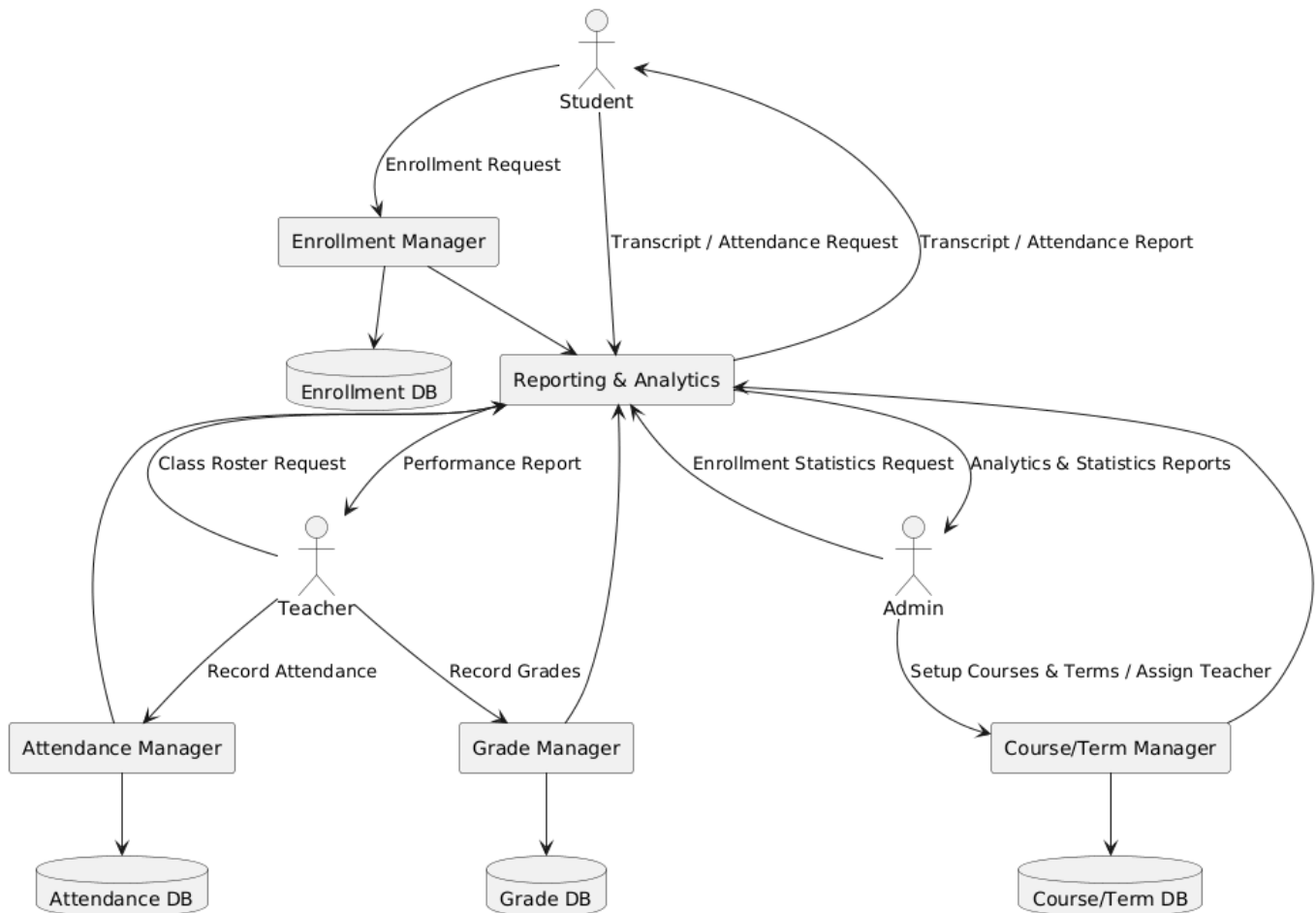
Level 0: Context Diagram



- **Student:** sends enrollment/record requests, receives transcripts and attendance
- **Teacher:** records grades/attendance, receives reports
- **Admin:** sets up courses/terms, receives analytics

Level 1: System Decomposition

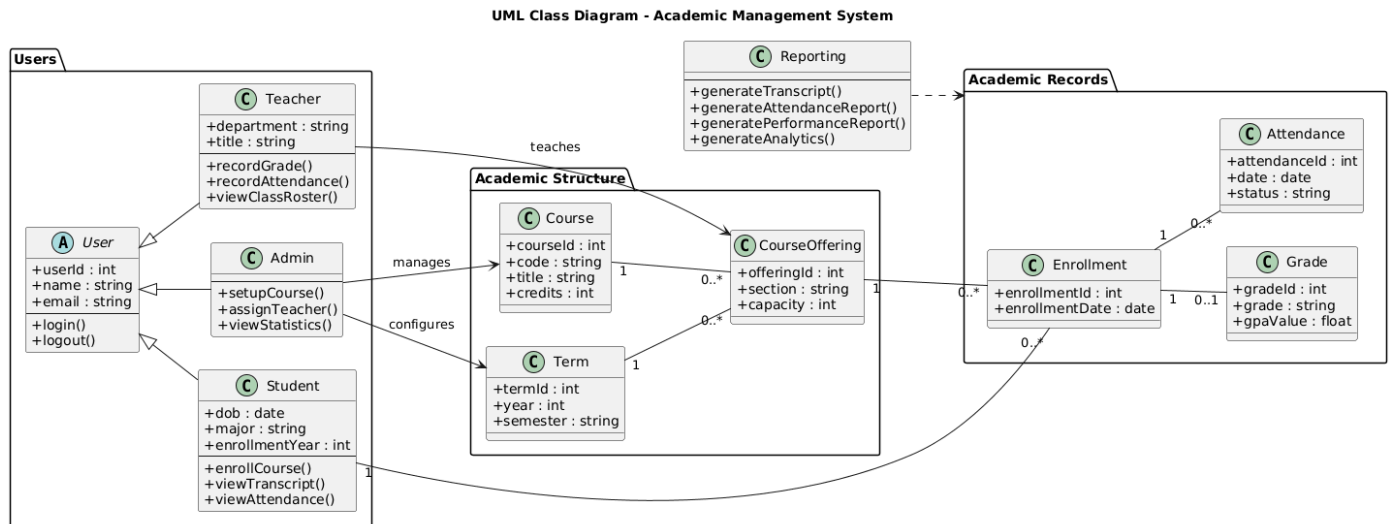
DFD - Level 1 (System Decomposition)



- **Enrollment Manager** → handles student enrollments
- **Grade Manager** → handles grade submissions
- **Attendance Manager** → records attendance
- **Course/Term Manager** → admin course setup and teacher assignments
- **Reporting & Analytics** → generates reports for all roles

5. UML Class Diagram

UML Diagram



Explanation

- **User (abstract)** → base class for Student, Teacher, Admin
- **Student** → can enroll in courses, view transcripts, view attendance
- **Teacher** → can record grades, record attendance, view class rosters
- **Admin** → can setup courses, assign teachers, view statistics
- **Course / Term / CourseOffering** → academic structure
- **Enrollment / Grade / Attendance** → academic records
- **Reporting** → generates transcripts, attendance reports, performance reports, analytics

6. Summary

- **SRS** defines the system's purpose, scope, and requirements.
- **Data Dictionary + ERD** define the **database schema**.
- **DFD** defines **data flow and process logic**.
- **UML** defines **system classes, methods, and relationships**.

Together, these artifacts provide a **complete design specification** for the Academic Management System.